Sequence 10

VARIATIONS AND EXTREMA OF A FUNCTION

1–12 February 2027

I Increasing, decreasing and monotonicity

Definitions

On an interval I, f is increasing if for all $a, b \in I$, $a < b \Rightarrow f(a) \leq f(b)$; decreasing if $a < b \Rightarrow f(a) \geq f(b)$. The variations are strict when the inequalities between images are strict. A constant function has the same value everywhere; it is both increasing and decreasing in the broad sense. A function that is monotonic on I is increasing or decreasing there.

Table of variations

Indicate the domain, the intervals of monotonicity, the values at the endpoints when defined, then the arrows. A sign table shows whether $f(x)$ is positive or negative; a table of variations shows how its values change.

Example 10.1

On $[-3; 4]$, $f(x) = (x-1)^2 - 2$. Using the square, extending the curve studied in sequence 6, draw its table of variations. Give two numbers $a < b$ such that $f(a) > f(b)$.

Solution:

Example 10.2

Can a function be negative and increasing? Give an example on an interval.

Solution:

II Maximum and minimum on an interval

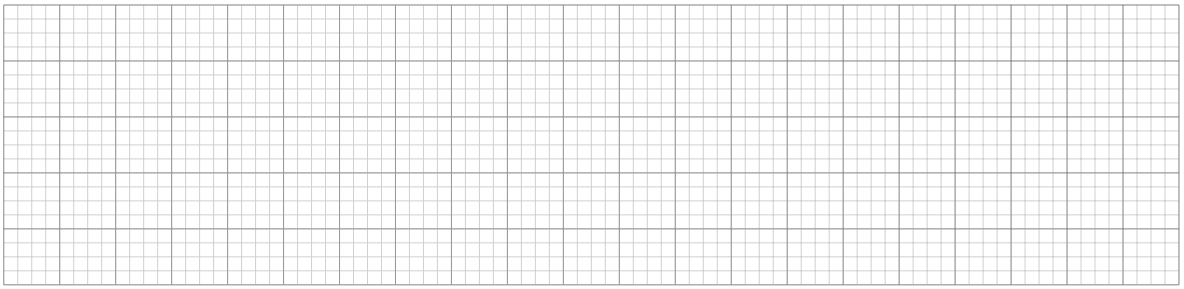
Definitions

f has a maximum at $a \in I$ if, for every $x \in I$, $f(x) \leq f(a)$. The value of the maximum is $f(a)$. f has a minimum at $b \in I$ if, for every $x \in I$, $f(x) \geq f(b)$. An extremum must be attained. Distinguish its value from the abscissa where it is attained.

Example 10.3

Return to $f(x) = (x-1)^2 - 2$ on $[-3; 4]$. Give the minimum and maximum, then where they are attained. Repeat on $[0; 4]$.

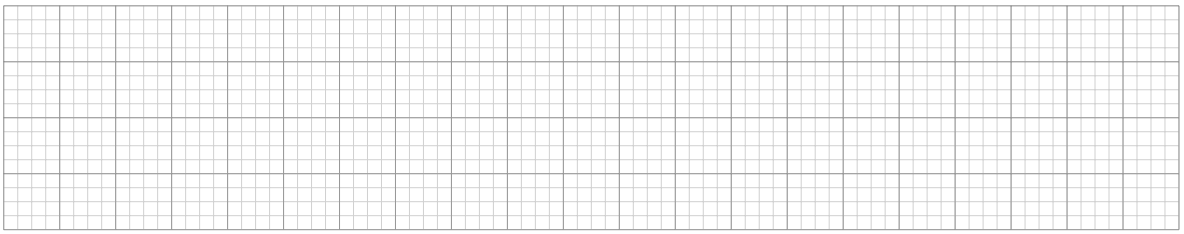
Solution:



Example 10.4

Does the function $f(x) = x$ have a maximum on $[0; 1[$? A minimum?

Solution:



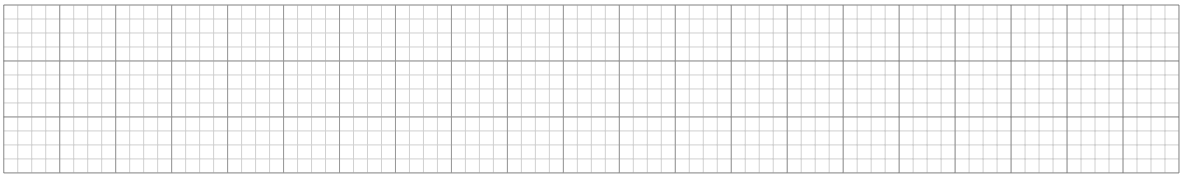
Reading without inventing

A table of variations does not provide all the intermediate images. It allows comparing certain images; if the abscissae lie on different branches, the comparison may be indeterminate.

Example 10.5

Does the mere fact that a function decreases then increases allow comparing two images on either side of the minimum?

Solution:



III Variations of affine functions

Property and rate of change

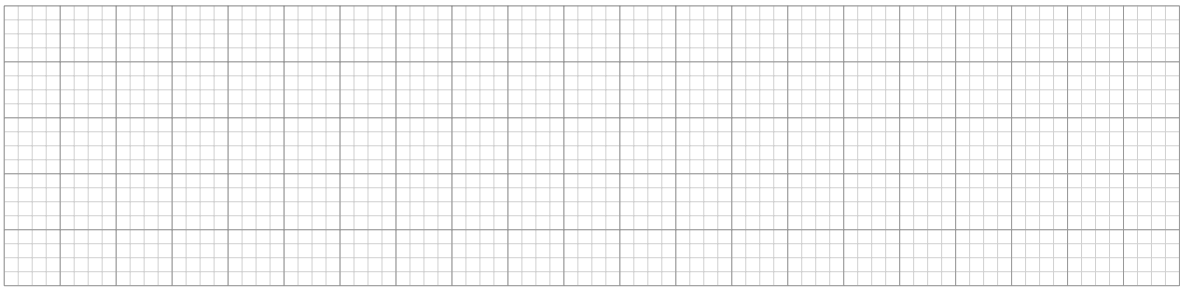
For $f(x) = mx + p$ and $a \neq b$,

$$\frac{f(b) - f(a)}{b - a} = m.$$

If $m > 0$, f is strictly increasing; if $m < 0$, strictly decreasing; if $m = 0$, constant.

Proof to write.

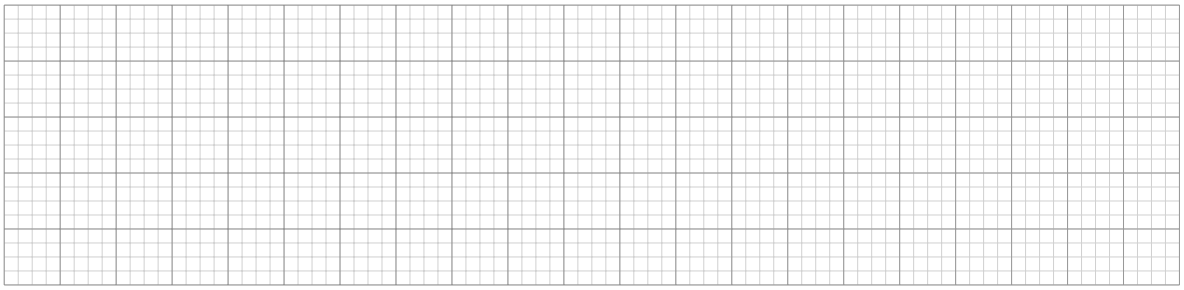
Solution:



Example 10.6

Determine the variations of $f(x) = 3x + 2$, $g(x) = 7 - 6x$, $h(x) = -x$. Give the sign of $2x - 6$ and explain why the sign of this function does not stay constant although it is increasing.

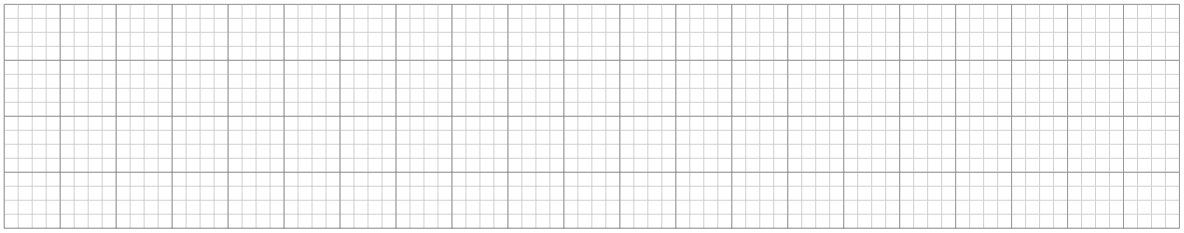
Solution:



Example 10.7

An affine function u satisfies $u(2) = 5$, $u(6) = 13$. Determine u, then its minimum and maximum on $[-1; 4]$.

Solution:



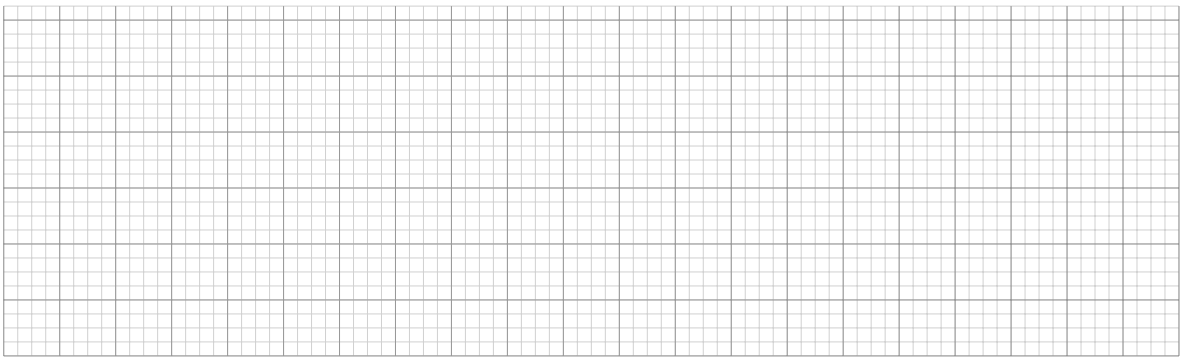
IV Variations of the square and the absolute value

Properties

$x \mapsto x^2$ and $x \mapsto |x|$ are strictly decreasing on $]-\infty;0]$ and strictly increasing on $[0;+\infty[$. Their minimum on $\mathbb{R}$ is 0, attained only at 0.

Proof to write.

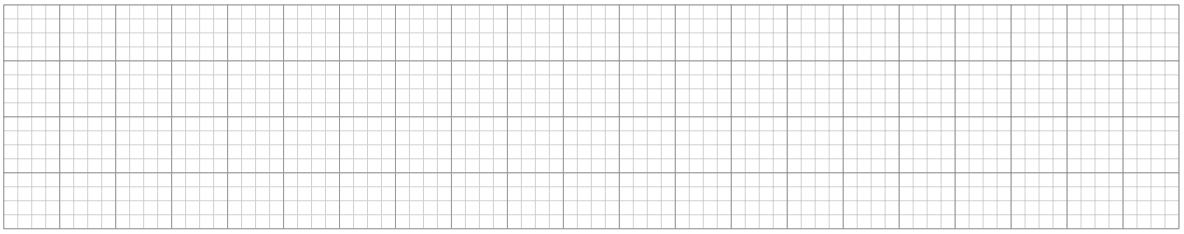
Solution:



Example 10.8

Compare without a calculator: $(-3.2)^2$ and $(-3.1)^2$; 1.2^2 and 1.3^2; $|-5|$ and $|-2|$.

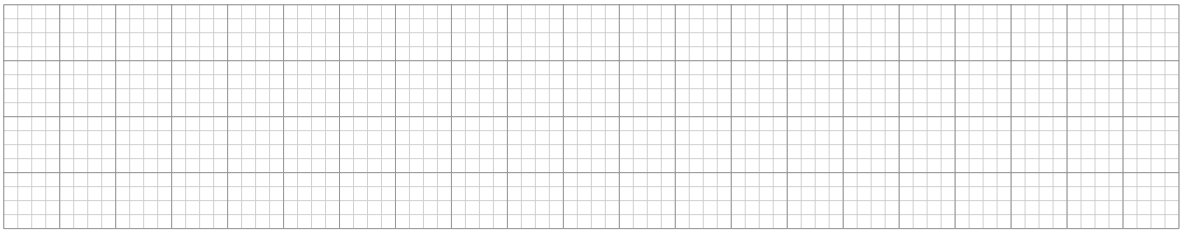
Solution:



Example 10.9

Bound x^2 if $-3 \leq x \leq 2$. Why can we not simply square the two bounds?

Solution:



V Variations of the inverse function

Property

The inverse function is strictly decreasing on $]-\infty;0[$ and on $]0;+\infty[$, separately. It is not decreasing on the whole of $\mathbb{R}^*$.

Proof to write.

Solution:

Example 10.10

Compare $1/(-4)$ and $1/(-2)$, then $1/3$ and $1/5$. Bound $1/x$ if $2 \leq x \leq 5$, then if $-5 \leq x \leq -2$.

Solution:

Example 10.11

Why is the interval $[-1;1]$ not suitable for directly applying a variation rule of the inverse function?

Solution:

VI Relative position of a number and its square

Property on the positives

For $0 \leq x \leq 1$, $x^2 \leq x$; for $x \geq 1$, $x^2 \geq x$. Equality holds at 0 and 1.

Proof to write.

Solution:

Square root and cube

The square root function is strictly increasing on $[0; +\infty[$. The cube function is strictly increasing on $\mathbb{R}$.

Example 10.12

Explain why the square root is increasing from the increase of the square on the positives. For the cube, use $b^3 - a^3 = (b-a)(a^2+ab+b^2)$.

Solution:

Example 10.13

Compare $\sqrt{5}$ and $\sqrt{6}$, then $(-2)^3$ and $(-1)^3$.

Solution:

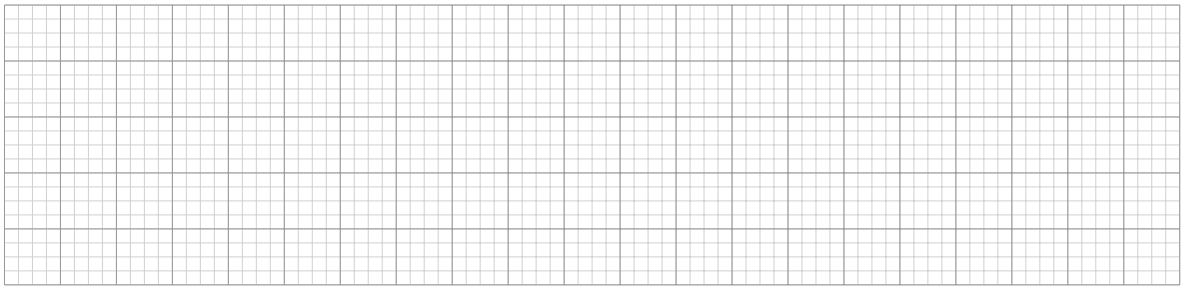
VII Optimising: conjecture and prove

Example 10.14

A rectangle of perimeter 20 m has area $A(x) = x(10 - x)$ with $x \in]0; 10[$.

1. Conjecture the maximum area using a table of values.
2. Show that $A(x) = 25 - (x - 5)^2$. Deduce the exact maximum and the corresponding dimensions.

Solution:



```
def aire(x):
    return x*(10-x)

def maximum_grille(N):
    # N entier >= 2 ; points strictement entre 0 et 10
    meilleur_x = 10/N
    meilleur_y = aire(meilleur_x)
    for k in range(2, N):
        x = 10*k/N
        if aire(x) > meilleur_y:
            meilleur_x, meilleur_y = x, aire(x)
    return meilleur_x, meilleur_y
```

Example 10.15

What does the program return for $N = 10$? Does a numerical sweep always prove the maximum of a function on an interval?

Solution:

